

Tanishq Sutrave

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EDUCATION

Mahindra University

June 2027

B.Tech in Computer Science and Engineering

Current GPA: 8.87

Coursework

AI & ML: NLP, Computer Vision, Reinforcement Learning, Generative AI (LLMs, RAG, Fine-tuning)

Math: Linear Algebra, Probability, Statistics, Optimization Techniques, Calculus

Core CS: Data Structures & Algorithms, OOP (Java), DBMS

TECHNICAL SKILLS

Machine Learning: PyTorch, Scikit-learn, OpenCV, TensorFlow, SHAP, LIME

Languages: Python, Java, SQL

Dev & Data Tools: Git/GitHub, Pandas, NumPy, Matplotlib, Jupyter, VS Code

PROJECTS

Hybrid Predictive Maintenance using Enhanced CMAPSS NASA Dataset

Apr 2025 – May 2025

Python, Machine Learning, Time-Series Analysis

- Designed a multi-stage predictive maintenance framework using the enhanced NASA CMAPSS turbofan engine dataset to model engine degradation and estimate Remaining Useful Life (RUL).
- Built a hybrid pipeline combining clustering to identify degradation stages, classification to label engine health states, and regression to predict time to next failure.
- Engineered temporal and sensor-derived features to capture performance drift across operational cycles.
- Developed a custom risk scoring mechanism integrating model outputs to prioritize maintenance actions and support data-driven maintenance planning.

Mobile Phone Price Range Classification

Jul 2025

Python, Scikit-learn, XGBoost, Pandas, Seaborn

- Built machine learning models on a dataset of 2000+ mobile phone specifications to classify devices into four price categories (0–3).
- Performed EDA and feature engineering (pixel resolution and screen area features) and applied preprocessing techniques including Isolation Forest for outlier detection and StandardScaler.
- Trained and tuned 8+ algorithms (Random Forest, SVM, XGBoost, Stacking), achieving **93.8% test accuracy** with Random Forest; analysis identified **RAM as the strongest predictor of price range**.

Energy Consumption Forecasting

Oct 2025

Python, Scikit-learn, Pandas

- Developed a predictive model to forecast time-series energy data, achieving a high R-Squared value.
- Implemented extensive feature engineering (lag and rolling statistics) to capture seasonality trends.

EXPERIENCE & ACHIEVEMENTS

Vice President

Aug 2025 – Present

Entrepreneurship & Innovation Cell, Mahindra University

- Led entrepreneurship initiatives by organising flagship events, startup talks, and innovation workshops engaging 100+ students.
- Helped build a strong campus startup culture through collaborative innovation programs.

Machine Learning Intern

Jul 2025 – Sep 2025

Unified Mentor

- Developed 4 end-to-end ML projects (heart disease prediction, mobile price classification, fraud detection, and ASL recognition) using Python, Scikit-learn, TensorFlow, and OpenCV, achieving up to **95% accuracy**.
- Built complete ML pipelines including data preprocessing, feature engineering, and model evaluation, applying algorithms such as Random Forest, SVM, Logistic Regression, and CNN for predictive analytics and real-time detection.

Webmaster – IEEE Computational Intelligence Society (CIS)

Feb 2026 – Present

IEEE Student Branch, Mahindra University

- Manage and maintain the digital presence of the IEEE CIS chapter, including website updates and publishing technical and event-related content.
- Collaborate with the CIS executive team to promote AI and computational intelligence events, increasing student engagement and visibility of chapter activities.

HONORS & AWARDS

- Academic Excellence: Awarded Academic Scholarship twice for maintaining high GPA.
- 1st Place: Quantum ML advanced-level hackathon (Team Quantum Intelligence).